

# CyberShield

Developed by Sarvan

Secure File Encryption and  
Password Strength Analysis System



Project by  
**Sarvan**

# CyberShield

**Developed by Sarvan**Secure File Encryption and Password Strength Analysis System

## Objective

To demonstrate secure authentication, password strength checking, file encryption/decryption and basic security scanning using Python.

## Problem Statement

Users often use weak passwords and store files without protection. This project demonstrates basic cybersecurity controls.

## Research Findings

Strong passwords, encryption, hashing and secure authentication significantly improve data protection.

## Methodology

Python, SQLite, SHA-256 password hashing, Fernet encryption, modular design and testing.

## Tools Used

Python, SQLite, Cryptography, Termux, VS Code/Nano.

## Implementation

Modules include Register/Login, Password Checker, Encrypt, Decrypt and Security Scan.



## Screenshots

Insert screenshots of Register, Login, Password Checker, Encryption, Decryption and Scan while demonstrating the project.

## Results

The system successfully registers users, validates login, checks password strength, encrypts/decrypts files and performs a simple file extension scan.

## Challenges

Package installation in Termux, dependency management and file permissions.

## Future Scope

GUI, malware scanning API, OTP login, cloud backup, report generation.

## Conclusion

CyberShield is a beginner-friendly cybersecurity project demonstrating essential security concepts.

## Sample Python Modules

- User Registration & Login using SQLite
- SHA-256 Password Hashing
- Password Strength Checker
- File Encryption/Decryption using Fernet
- Security Scanner

```

No mirror or mirror group selected. You might want to select one by running 'termux-change-repo'
Checking availability of current mirror:
[*] https://termux.danyael.xyz/termux/termux-main: ok
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  binutils python-cryptography
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 3564 kB of archives.
After this operation, 25.8 MB of additional disk space will be used.
Get:1 https://termux.danyael.xyz/termux/termux-main stable/main aarch64 binutils aarch64 2.46.0-4 [2437 kB]
Get:2 https://termux.danyael.xyz/termux/termux-main stable/main aarch64 python-cryptography aarch64 48.0.1-2 [1127 kB]
Fetched 3564 kB in 10s (361 kB/s)
Selecting previously unselected package binutils.
(Reading database ... 18841 files and directories currently installed.)
Preparing to unpack .../binutils_2.46.0-4_aarch64.deb ...
Unpacking binutils (2.46.0-4) ...
Selecting previously unselected package python-cryptography.
Preparing to unpack .../python-cryptography_48.0.1-2_aarch64.deb ...
Unpacking python-cryptography (48.0.1-2) ...
Setting up python-cryptography (48.0.1-2) ...
Installing dependencies for python-cryptography through pip...
Requirement already satisfied: cffi>=1.12 in ./data/data/com.termux/files/usr/lib/python3.14/site-packages (2.0.0)
Requirement already satisfied: cryptography in ./data/data/com.termux/files/usr/lib/python3.14/site-packages (48.0.1)
Requirement already satisfied: pycparser in ./data/data/com.termux/files/usr/lib/python3.14/site-packages (from cffi>=1.12) (3.0)
Setting up binutils (2.46.0-4) ...
~/CyberShield $ python CyberShield.py

```

```

===== CyberShield =====

```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

```

Enter Choice:

```

```

Invalid Choice

```

```

===== CyberShield =====

```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

```

Enter Choice: █

```

```

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```

===== CyberShield =====

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 2

Username: @sanju

Password: 123456789

Login Successful

===== CyberShield =====

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 3

Enter Password: 123456789

Weak Password

===== CyberShield =====

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 4

File Name: 6

Error: [Errno 2] No such file or directory: '6'

===== CyberShield =====

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 6

Enter File Name: python CyberShield.py

File appears safe.

===== CyberShield =====

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: █

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eb ...
Unpacking python-cryptography (48.0.1-2) ...
Setting up python-cryptography (48.0.1-2) ...
Installing dependencies for python-cryptography through pip...
Requirement already satisfied: cffi>=1.12 in ./data/data/com.termux/files/usr/lib/python3.14/site-packages (2.0.0)
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```

```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice:

Invalid Choice

```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 1

Username: @sanju

Password: 123456789

Registration Successful

```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: 2

Username: @sanju

Password: 123456789

Login Successful

```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

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```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice:

Invalid Choice

```
===== CyberShield =====
```

1. Register
2. Login
3. Password Checker
4. Encrypt File
5. Decrypt File
6. Security Scan
7. Exit

Enter Choice: █

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